# UN OpenGIS WG5
# GEOAI and Analytics
## Monthly Report



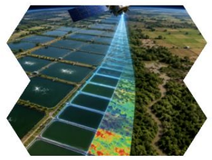
# Contents

- GEO-AI Challenge 2026
- Agro-Informatics Digital Assistant (AIDA)
- FERSPAS

ZINDI

Compete Jobs Learn Chat Leaderboard More

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## GeoAI Aquaculture Pond Identification Challenge by FAO and ITU

**1000 CHF**

2 months left

Skills you will learn: Data Analysis, Classification, Feature Engineering, GIS

527 joined 163 active

Info Data Chat Leaderboard

Join

| Start | Enrolments close | Close | Reveal |
|---|---|---|---|
| 08 Jun 26 | 07 Aug 26 | 16 Aug 26 | 17 Aug 26 |

### Can you identify aquaculture ponds from space?

Aquaculture ponds are used to farm fish and shrimp in controlled water environments. They are easier to manage and more productive than open-water fishing. But to monitor them effectively, we first need to know where they are. Satellite imagery offers a reliable way to map these ponds across large areas, providing up-to-date information that supports water management, environmental monitoring, and policy decisions, especially in regions where ground-level data is hard to come by.

Your goal is to build a machine learning model that predicts whether a given location is an aquaculture pond or another type of land cover, using features derived from satellite data. Each location represents a 10m × 10m patch of ground, so your model needs to be precise.

There is an added challenge: your model will be trained on data from one time period and tested on data from a different one. This means your solution must remain accurate as conditions change across seasons and years — not just perform well on familiar data.

By taking part in this challenge, you will contribute to global efforts in agricultural land mapping and support progress towards the United Nations Sustainable Development Goals (SDGs) for 2030.

ITU

FAO FIAT PANIS

Food and Agriculture Organization of the United Nations

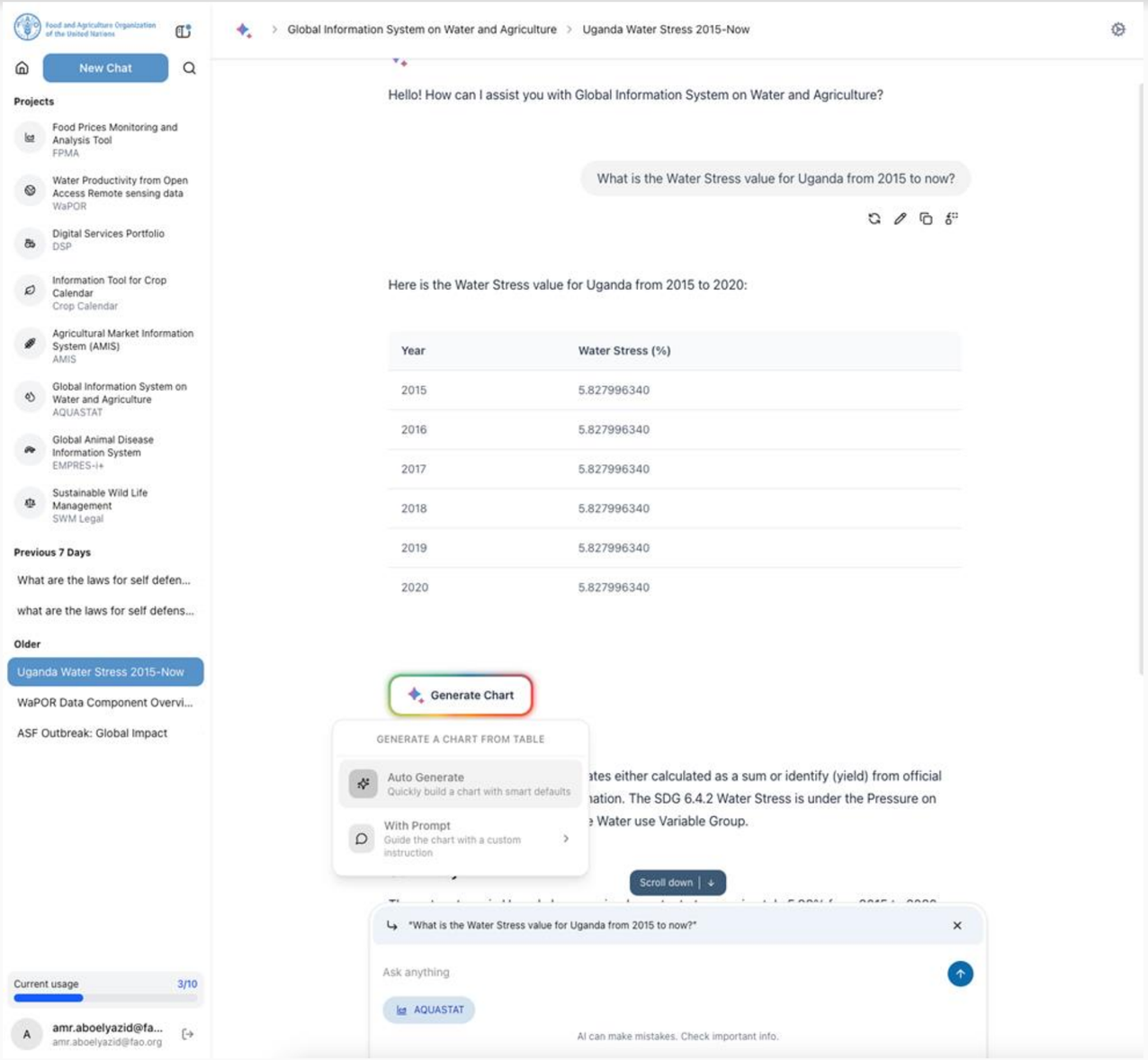
https://zindi.africa/competitions/geoai-aquaculture-pond-identification-challenge

# Agro-Informatics Digital Assistant (AIDA)

AIDA is an intelligent conversational interface designed to enhance user interaction with the Agro-informatics platform grounded in agro-informatics data.

Easily query project documentation, retrieve live data, and interact with complex tools through a simple chat interface.

- Single domain or or multi-agent support
- Retrieval-Augmented Generation (RAG): answers
- Agents capable of function Calling to fetch live data from the Platform
- AI Report Generation




https://aida.fao.org/

# https://data.fao.org

Search for locations

Explore Data | Upload

Map

UN Country Boundaries of the World

Place Labels

X

## Food and Agriculture Organization of the United Nations

## Welcome to the FAO Agro-informatics Platform




The FAO Agro-informatics Platform (AIP) is a part of FAO digital infrastructure and one stop shop digital platform for member countries, organizations, stakeholders, and researchers to access data, knowledge, and solutions in agrifood systems—supporting efforts to enhance productivity, profitability, and sustainability, reduce poverty, end hunger, and promote sustainable development. Powered by cutting-edge technologies such as Artificial Intelligence (AI), big data, geospatial informatics, and cloud computing, AIP provides advanced digital tools for targeted, data-driven analysis and interventions at various scales — from farm, region, country, continent, to the globe. The platform enables users to turn data and knowledge into practical insights for policy design, investment planning, and multiple-scale action toward a food-secure, sustainable future.

Take a Quick Tour | Start Exploring | Read More

Close and Don't Show Again

Give Feedback

© FAO Disclaimer | Leaflet | © MapTiler © OpenStreetMap contributors

Chats

Search... ⌘K

Create a folder

TODAY

Temperature growing se...

show me a map of temperature growing period

Thought for 2s

Complete

Several temperature growing period datasets are available! Please choose the one that best suits your needs:

Dataset Selection
Days with Ta > 30 °C (Annual, ~10 km) – GAEZ v5

Loading Map Layers
Retrieving matching map layers...

Balanced

Ask anything...

K



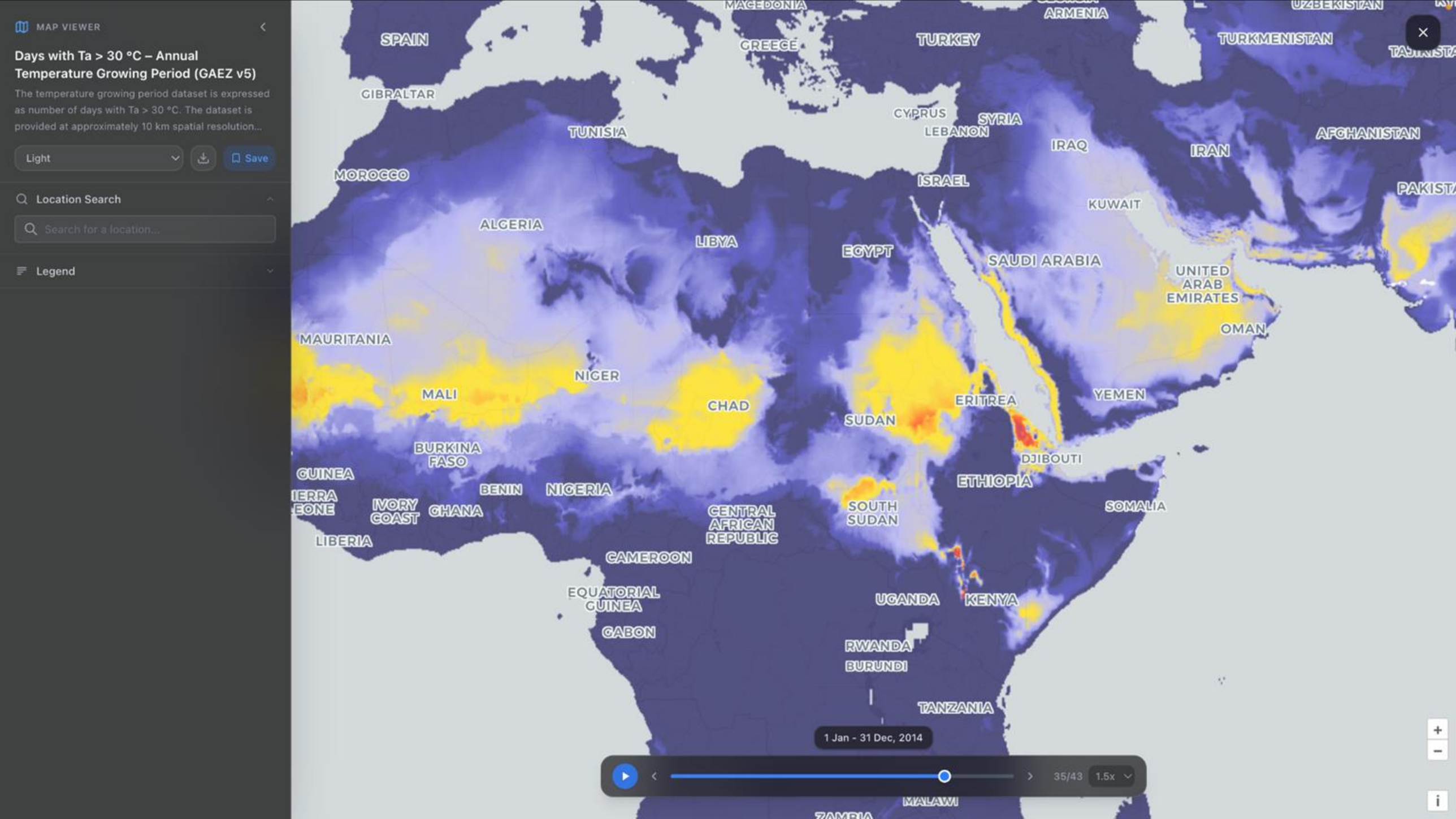

Here is the global map showing the **annual number of days with mean air temperature above 30°C**, sourced from the **GAEZ v5** dataset (Global Agro-Ecological Zones, Version 5).

## Key Details:

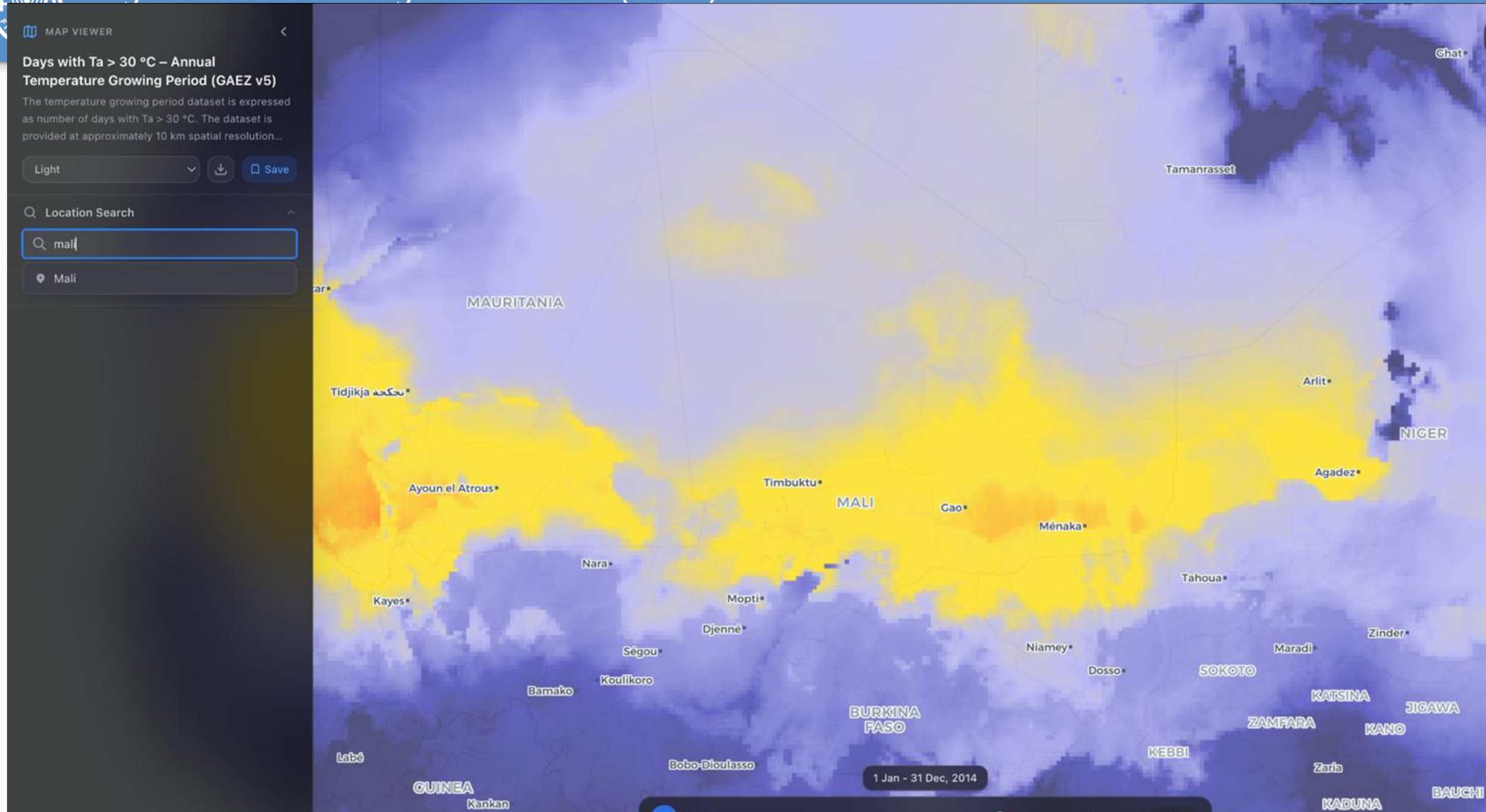
- **Spatial Resolution:** ~10 km
- Balanced **overage:** 1980–2022 (annual time series)

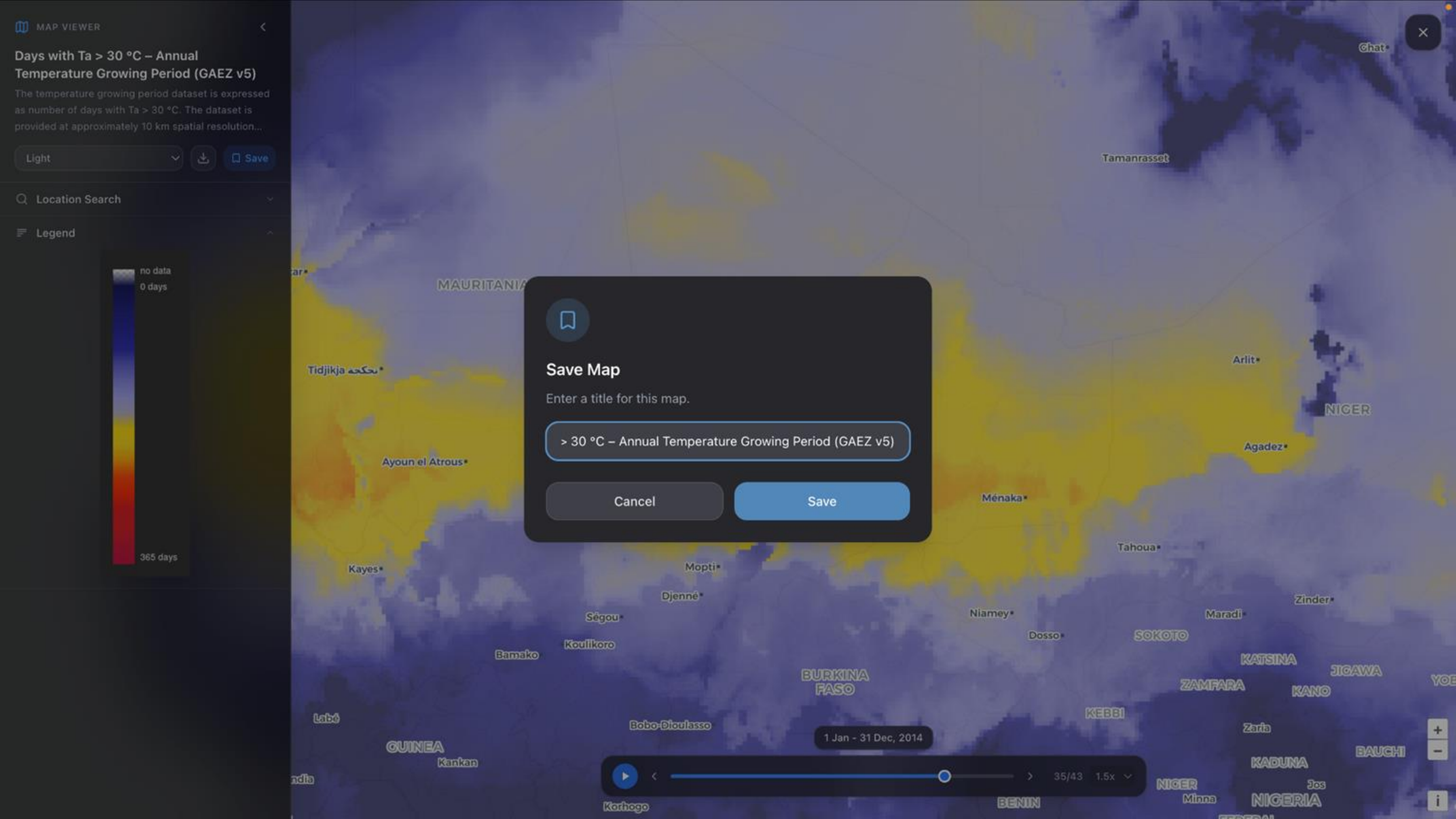
Ask anything...

- **Climate Source:** AgERA5 data from the Copernicus Programme


MAP VIEWER
Days with Ta > 30 °C – Annual Temperature Growing Period (GAEZ v5)
The temperature growing period dataset is expressed as number of days with Ta > 30 °C. The dataset is provided at approximately 10 km spatial resolution...
Light
Save
Location Search
Search for a location...
Legend
SPAIN
GIBRALTAR
MOROCCO
TUNISIA
ALGERIA
LIBYA
EGYPT
MAURITANIA
NIGER
MALI
CHAD
SUDAN
ERITREA
BURKINA FASO
GUINEA
BENIN
NIGERIA
IVORY COAST
GHANA
LIBERIA
CENTRAL AFRICAN REPUBLIC
SOUTH SUDAN
CAMEROON
EQUATORIAL GUINEA
GABON
UGANDA
KENYA
RWANDA
BURUNDI
TANZANIA
ETHIOPIA
SOMALIA
DJIBOUTI
YEMEN
OMAN
SAUDI ARABIA
UNITED ARAB EMIRATES
KUWAIT
IRAQ
IRAN
SYRIA
LEBANON
CYPRUS
ISRAEL
TURKEY
GREECE
ARMENIA
TURKMENISTAN
AFGHANISTAN
1 Jan - 31 Dec, 2014
35/43
1.5x

# Agro-Informatics Digital Assistant (AIDA)




MAP VIEWER
Days with Ta > 30 °C – Annual Temperature Growing Period (GAEZ v5)
The temperature growing period dataset is expressed as number of days with Ta > 30 °C. The dataset is provided at approximately 10 km spatial resolution...
Light
Save
Location Search
Legend
no data
0 days
365 days
Save Map
Enter a title for this map.
> 30 °C – Annual Temperature Growing Period (GAEZ v5)
Cancel
Save
1 Jan - 31 Dec, 2014
35/43
1.5x
Ghat
Tamanrasset
Arlit
Agadez
NIGER
Tidjikja
Ayoun el Atrous
Ménaka
Tahoua
Kayes
Mopti
Djenné
Ségou
Niamey
Zinder
Maradi
Dosso
SOKOTO
Koulikoro
Bamako
KATSINA
JIGAWA
ZAMFARA
KANO
BURKINA FASO
KEBBI
Labé
Bobo-Dioulasso
Zaria
GUINEA
Kankan
KADUNA
BAUCHI
Jos
Minna
NIGERIA
Korhogo
BENIN

Here is the global map showing the **annual number of days with mean air temperature above 30°C**, sourced from the **GAEZ v5** dataset (Global Agro-Ecological Zones, Version 5).

### Key Details:

- **Spatial Resolution:** ~10 km
- **Temporal Coverage:** 1980–2022 (annual time series)
- **Current Display:** Year 2022 (you can use the time selector to browse other years)
- **Climate Source:** AgERA5 data from the Copernicus Programme

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## Summary

This map highlights regions globally where mean daily temperatures exceed **30°C**, a critical threshold for many crops. Areas in **tropical Africa, South Asia, the Middle East, and parts of Latin America** typically show the highest number of such days per year, which can be indicative of heat stress on crops. Higher-latitude and high-altitude regions tend to show very few or zero such days. The annual time series (1980–2022) allows users to observe **long-term warming trends** over time.

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Opens the link in a new tab: https://www.fao.org/gaez/en/
Open Link
Open Link
Balanced
Ask anything...



Search... ⌘K

Create a folder

TODAY

- Maize production in Mali
- Definition of Gaez
- Temperature growing sea...

## Maize Production in Mali – Last 10 Y...

10 rows · 4 columns

Save | Explore

| YEAR | AREA HARVESTED (HA... | PRODUCTION (TONNES... | YIELD (HG/HA) |
|---|---|---|---|
| 2015 | 899,640 | 2,276,036 | 2,529.9 |
| 2016 | 1,031,522 | 2,811,385 | 2,725.5 |
| 2017 | 1,233,008 | 3,598,205 | 2,918.2 |
| 2018 | 1,258,559 | 3,624,950 | 2,880.2 |
| 2019 | 1,432,151 | 3,816,536 | 2,664.9 |
| 2020 | 1,170,148 | 3,516,865 | 3,005.5 |
| 2021 | 1,536,189 | 3,603,000 | 2,345.4 |
| 2022 | 1,454,000 | 3,904,687 | 2,685.5 |

2 more rows · Total 10

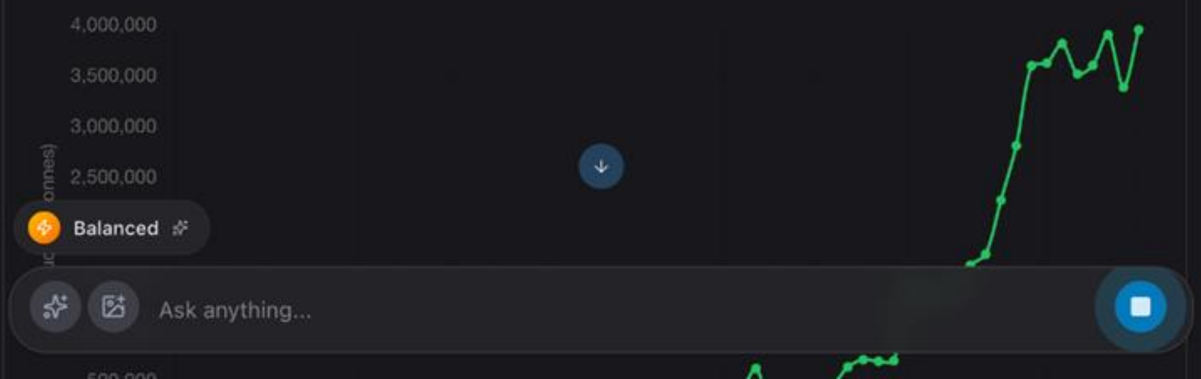
Sources (1) DIRECT

Created just now

## Maize Production in Mali (1961–2024)

Line Chart › 64 data points

Save



Chats

Search... ⌘K

Create a folder

TODAY

Maize production in Mali

Definition of Gaez

Temperature growing sea...

Sources (1) DIRECT

Created just now

## Maize Production in Mali (1961–2024)

Line Chart › 64 data points

Save

Save chart



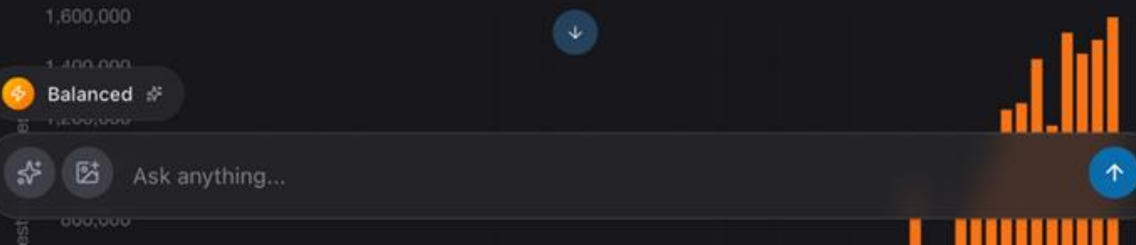

Sources (1) DIRECT

Created just now

## Maize Area Harvested in Mali (1961–2024)

Bar Chart › 64 data points

Save




Balanced

Ask anything...

Workspaces

Mali Maize
1 page · 1m ago

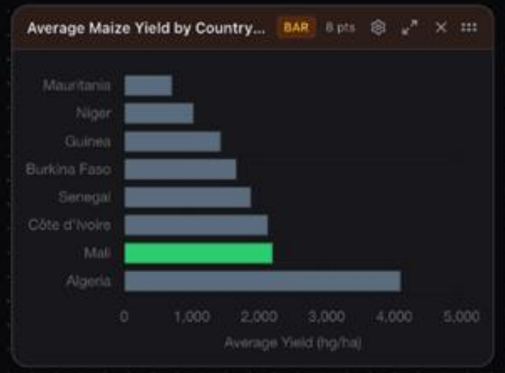
Mali Maize

Export Share

Page 1

Layers







Charts

Search charts...

Average Maize Yield by Cou...
Chart

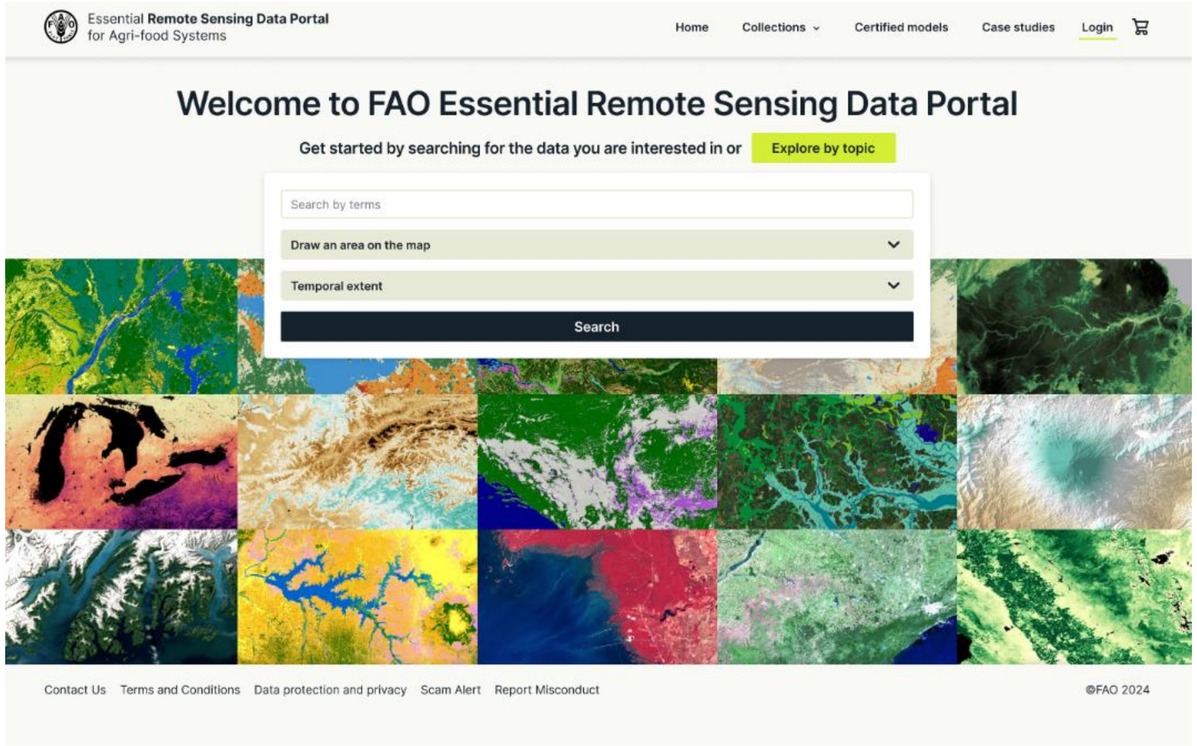
Maize Yield Trend in Mali vs ...
Chart

Maize Production in Mali (19...
Chart

75%

# Essential Remote Sensing Data Product Portal

- Portal for User acquiring datasets
  - Scope: Provide **certified** up-to-date and ready-to-analysis remote sensing product and reference data for validation.
  - Target User: Decision maker and Data analysts who using satellite data product to assist their decision and analysis.
  - Data Policy: Open access with data citation.




https://data.review.fao.org/remote-sensing-portal

# Essential Remote Sensing Data Product Portal

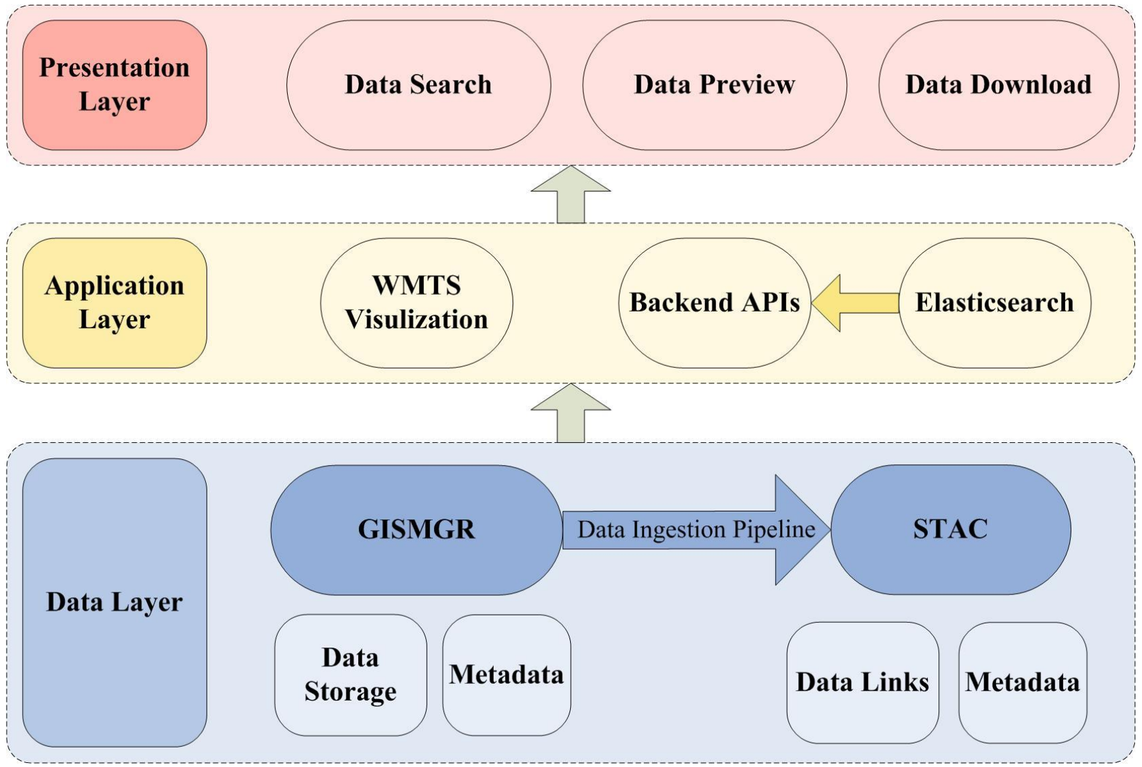
- System Architecture

➢ Data layer: All remote sensing and metadata are currently organized by GISMGR 2

➢ Communication Layer: The Data are ingested to STAC, which is the basement of all the search and download services

➢ Communication Layer: All data search APIs are developed based on the STAC, and data visualization APIs are created based on WMTS

➢ Presentation Layer: Frontend providing services to users




Portal Architecture

# Essential Remote Sensing Data Product Portal

- Major Functions
  - ➢ Data Catalog
  - ➢ Data Search
  - ➢ Data Download

